

Sleep and Sleep Disorders 4



Sleep function: an evolutionary perspective

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Prospective epidemiological studies in industrial societies indicate that 7 h of sleep per night in people aged 18 years or older is optimum, with higher and lower amounts of sleep predicting a shorter lifespan. Humans living a hunter-gatherer lifestyle (eg, tribal groups) sleep for 6–8 h per night, with the longest sleep durations in winter. The prevalence of insomnia in hunter-gatherer populations is low (around 2%) compared with the prevalence of insomnia in industrial societies (around 10–30%). Sleep deprivation studies, which are done to gain insights into sleep function, are often confounded by the effects of stress. Consideration of the duration of spontaneous daily sleep across species of mammals, which ranges from 2 h to 20 h, can provide important insights into sleep function without the stress of deprivation. Sleep duration is not related to brain size or cognitive ability. Rather, sleep duration across species is associated with their ecological niche and feeding requirements, indicating a role for wake–sleep balance in food acquisition and energy conservation. Brain temperature drops from waking levels during non-rapid eye movement (non-REM) sleep and rises during REM sleep. Average daily REM sleep time of homeotherm orders is negatively correlated with average body and brain temperature, with the largest amount of REM sleep in egg laying (monotreme) mammals, moderate amounts in pouched (marsupial) mammals, lower amounts in placental mammals, and the lowest amounts in birds. REM sleep might, therefore, have a key role in the regulation of temperature and metabolism of the brain during sleep and in the facilitation of alert awakening.

Introduction

Under natural conditions, most diurnal non-human primates go to sleep near sunset and awaken around dawn, sleeping for about 10–12 h.¹ Human beings might be expected to show this same pattern of dusk-to-dawn sleep. However, a 2015 study of hunter-gatherer groups living in a traditional manner in natural environments in Africa and South America has refuted this assumption.² This study showed that these populations do not sleep for 10–12 h as do non-human primates; rather, these people sleep for roughly 6–8 h. Another expectation about human sleep is that the artificial environments and electric lighting to which industrial populations (ie, people living in societies driven by technology) are exposed might greatly reduce sleep duration by delaying sleep onset. Although insomnia is much more prevalent in industrial populations than in humans living a hunter-gatherer lifestyle,² the common assumption that sleep duration in industrial societies has declined over the past 50 years or more has been contradicted by the findings of a literature review of studies of sleep duration in healthy adults.³

Sleep duration in humans varies by age, with the longest periods in newborn babies (12–15 h), an intermediate duration in teenagers (9–10 h), and a lower amount of sleep in adulthood (6–8 h). Healthy older people generally maintain this duration of sleep.⁴ Racial, ethnic, and sex differences in sleep duration have been noted in studies, but the causes of these variations remain to be determined.⁵

In human adults, both short and long durations of sleep (ie, shorter or longer than 7 h) are associated with poorer health outcomes than a 7 h duration. Epidemiological studies in industrial societies, including as many as 1 million participants who were followed up for a

maximum of 6 years, have consistently shown that 7 h of sleep predicts the longest lifespan.^{6–10} Pre-existing conditions such as sleep apnoea (which disrupts sleep and extends sleep duration), were controlled for in these studies. A striking finding was that, on average, lifespan was reduced in people sleeping more than 7 h compared with those sleeping less than 7 h.^{6–11} No evidence has suggested that sleep restriction in people sleeping longer than 7 h is beneficial. More sleep is not correlated with better health perhaps because sleep is incompatible with activity and exercise, which are well known to promote health.¹² A study of brain grey matter volume, white matter microstructure, and cognitive changes over 28 years in 613 participants of the Whitehall II cohort study found no correlation of these factors with sleep duration.¹³ Studies of the duration of spontaneous daily sleep across species of mammal, which can vary from 2 h to 20 h, can provide important insights into sleep function.

In this Review, I discuss the associations between sleep duration, health, and cognition in relation to modern industrial environments. Moreover, I present hypotheses on the function of rapid eye movement (REM) and non-REM sleep across species. This Review is part of a Series of four papers on sleep. The three other papers in the Series are published in *The Lancet* and discuss excessive daytime sleepiness,¹⁴ insomnia,¹⁵ and circadian rhythm disorders.¹⁶

Effects of the modern environment on sleep

The sleep patterns of humans living a hunter-gatherer lifestyle have been analysed as a means to determine the patterns of human sleep in environments similar to those in which humans evolved—ie, in times before the changes in living conditions brought about by industrial society.

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This is the fourth in a **Series** of four papers about sleep and sleep disorders (papers 1, 2 and 3 appear in *The Lancet*). All papers in the Series are available at www.thelancet.com/series/sleep-disorders

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Two tribal groups living in Africa (Namibia and Tanzania) and one group living in South America (Bolivia)² have been studied. All three of these groups have no electricity, minimal shelters, and are self-sufficient. These groups gather food and have little contact with industrial societies. Activity monitors that do not disturb sleep and have been validated with EEG recording were used to measure sleep duration and timing. All three groups exhibited similar sleep patterns. Contrary to assumptions that humans living a hunter-gatherer lifestyle sleep from dusk to dawn, individuals in the three tribal groups rarely went to sleep at sunset, with sleep onset occurring (on average) 3·2 h after sunset. The time between sunset and sleep onset was filled with conversation, socialising, and cooking over small fires, with a light intensity (ie, from the fires) generally below levels shown to alter circadian rhythms.¹⁷ These activities are somewhat similar to those occurring in industrialised societies. Indeed, it seems that being awake after sunset is a characteristic that distinguishes human beings from other diurnal primates.¹² The studies also showed that, contrary to some speculations about natural human sleep, hunter-gatherers do not regularly awaken in the middle of the night or nap at midday. However, these behaviours do occasionally occur,² as they do in industrial societies.

Many species studied in their natural environment have been shown to have pronounced seasonal differences in sleep duration.^{18–20} We found that the hunter-gatherer populations sleep for nearly 1 h longer in winter than in summer.² Hunter-gatherers sleep during the coldest night-time period, which starts more than 3 h after sunset and ends when the lowest daily temperatures are reached (ie, around dawn). Sleep onset time is much more variable than is sleep offset (ie, awakening) time. This natural situation is in keeping with traditional clinical advice to assist sleep maintenance by lowering night-time temperatures and to awaken at a fixed hour. Hunter-gatherers show peripheral vasoconstriction on awakening in the cold morning,² unlike most populations living in industrial societies.

Less than 2% of hunter-gatherers in Namibia and Bolivia reported or showed signs of insomnia, compared with the 10–30% rate of insomnia (depending on the defining criteria) recorded in industrial societies.^{21,22} Widespread insomnia is, thus, not a natural characteristic of human beings. The 6–8 h sleep duration of hunter-gatherer groups, which is slightly shorter than the 7–8 h or more duration that has been promoted as a goal in industrial societies,²³ is not incompatible with excellent health. For example, the Tsimane hunter-gatherer group in Bolivia has the lowest rate of coronary artery disease of any studied human population.²⁴ Hypertension is also rare in hunter-gatherer populations,²⁵ and prostate enlargement is virtually unknown.²⁶ Some evidence suggests that Alzheimer's disease could be relatively uncommon.²⁷ Obesity and diabetes are also quite rare in hunter-gatherer populations,^{28–30} which accords with

evidence that food technology in industrial societies has overwhelmed normal human appetite-regulating mechanisms.³¹ Hunter-gatherers who move to cities exhibit marked obesity, even exceeding the rates of obesity in industrial societies within a single generation.³² Because of the low rate of childhood vaccinations in groups living a hunter-gatherer lifestyle, child mortality is high, although vaccination rates have been increasing. For hunter-gatherers who survive childhood, their lifespan is similar to that reported for people living in industrialised societies.²⁴ The extent to which these features of hunter-gatherer health are dependent on diet, exercise, psychological factors associated with group living, or other variables remains to be determined.

Evolutionary determinants of sleep

Sleep is sometimes said to be a maladaptive state because it increases the vulnerability of animals to predation and is incompatible with eating, reproducing, and protecting offspring. However, the balance between energy acquisition and expenditure is a key determinant of evolutionary fitness and, hence, reproduction. Because energy expenditure is decreased by inactivity, including during hibernation,³³ estivation (ie, extended periods of dormancy during a hot or dry period), and sleep, these states increase evolutionary fitness. For example, some species of hummingbirds show nightly torpor, a state of greatly reduced body temperature that occurs after the animal goes to sleep. Body temperature can drop from 41°C to as low as 4°C during torpor in several species of Andean hummingbirds.³⁴ Nightly torpor has been interpreted as an energy-saving adaptation.³⁴ In mammals, body temperature and energy expenditure decrease during sleep,^{35,36} and energy acquisition requires waking. Elephants and other large herbivores maximise their reproductive success by achieving a small but reliable energy surplus through eating large quantities of vegetation. This consumption is directly proportional to the time these animals spend being awake.³⁷ The optimal balance between activity and inactivity varies by season, and depends on the diet and food availability of each species.

Although energy balance and ecological niches appear to be the main drivers of sleep,^{35,38–41} several other ideas have been proposed to explain why mammals sleep. One hypothesis from 2013 is that sleep might enable drainage of toxins from the brain.⁴² The evidence for this notion is that the brain's so-called glymphatic system functions better when the brain is relatively inactive (eg, during non-REM sleep and anaesthesia) than when the brain is active.⁴² But non-REM sleep is followed by REM sleep in almost all mammals. REM sleep is a state of very high brain metabolic activity that is similar to waking.⁴³ Therefore, toxin clearance in non-REM sleep would be undone by the re-creation of toxins in REM sleep. Whether there is a net clearance of toxins across the entire natural sleep period in all species remains to be

established. Another hypothesis of sleep function is that so-called excess synapses are removed during non-REM sleep. This process is specific for brain region and age.⁴⁴ The use of sleep deprivation in studies aiming to determine the function of sleep (eg, to test the synapse removal hypothesis) complicates their interpretation because stress is induced by the experimenter, which is a confounding factor (panels 1, 2).^{45–57} Both synapse removal and toxin drainage might occur during sleep, but neither can yet explain the enormous variation in sleep amount across species. Examples of such variation across species are: the differences in sleep duration between carnivores, omnivores, and herbivores;^{38,50} the absence of distinctive qualities of human sleep compared with other mammals, despite our high brain-to-bodyweight ratio and unique cognitive abilities;⁵⁰ and the large seasonal and migratory changes in sleep duration seen in many species, including humans.² These differences can be explained by energy conservation benefits and ecological considerations.

Sleep function in homeotherms

Analysis of the correlates of sleep duration across mammals has been used to facilitate understanding of the functional role of sleep. By contrast with sleep deprivation studies, analysing the correlates of sleep duration is not confounded by experimenter-induced stress. In general, species with longer durations of sleep appear to have deeper sleep (ie, the animal is harder to arouse), which is often characterised by the presence of high voltage non-REM sleep (as shown on an EEG) and elevated arousal thresholds to auditory or tactile stimuli. Most mammals, including humans, sleep for longer periods and have deeper sleep when they are young than when they are mature, although this is not the case with cetacean mammals (eg, whales and dolphins). Do animals that sleep for a longer period have larger brain-to-bodyweight ratios? Do smaller animals, with higher mass-specific metabolic rates and higher rates of free-radical generation, have longer sleep durations than animals with lower mass-specific metabolic rates? Does the lifespan, litter size, or body temperature across species correlate with non-REM or REM sleep duration? These research questions have been studied across species.

No substantial correlation between brain size or brain-to-bodyweight ratio and sleep duration has been reported across species. An inverse correlation between body mass and daily sleep duration was reported in early studies of sleep phylogeny.⁵⁸ This finding is a result of large herbivore species (eg, elephants, giraffes, and cattle) having very low amounts of sleep compared with omnivores and carnivores, which is consistent with large herbivores having evolved to spend most of their time awake and eating low-calorie food. However, sleep in carnivores and omnivores does not show a correlation with body mass (figure 1).³⁸ Across species, in studies done under laboratory or zoo conditions where food is freely supplied (ie, ad-lib), carnivores sleep more than omnivores, and

Panel 1: Limitations of studies using sleep deprivation

A traditional approach to determining the function of sleep is to deprive animals of sleep and attribute any resulting deficits to the loss of sleep function. This approach is problematic because of the stress involved in sleep deprivation. Deprivation typically involves repeated arousal of the studied animal as the animal drifts into sleep. Such arousals generate the cortisol awakening response.^{45,46} Awakening is associated with rapid (burst) firing of neurons in most brain regions.⁴³ The disk-over-water technique (developed for sleep deprivation studies in rats), although appearing gentle, results in over 1000 awakenings per day.^{47–49} Although rats die after 2 or more weeks of this procedure,⁵² the neuronal and systematic stress resulting from such awakenings cannot be separated from the loss of sleep time. Similarly, sleep deprivation by gentle handling, often used in sleep deprivation studies in mice, requires repeated arousals as animals drift into sleep. Experimental sleep deprivation in humans might qualitatively differ from animal sleep deprivation studies because the human participant volunteers for the procedure. Nevertheless, arousals from the sleepy state or from brief sleep onsets interrupted by the experimenter can be expected to repeatedly generate the physiological awakening response, which might, by itself, cause the induced symptoms.

Panel 2: Sleep and memory consolidation

An important variable in experimental studies on sleep and memory consolidation that has not been adequately controlled for is interference. Sleep deprivation might diminish recall, not by preventing memory consolidation but rather by increasing waking and its accompanying learning, which competes for consolidation with the learning measured by the experimenter,^{54–56} and by inducing stress. Relaxed quiet waking can be as effective as sleep in promoting recall of items presented by the experimenter.^{50–56} A study of the relation between REM sleep, non-REM sleep, other sleep parameters, and memory consolidation in more than 900 participants did not find any correlation between the sleep parameters and memory consolidation.⁵⁷ Another study with over 900 participants on intra-individual differences in sleep parameters and recall also found no correlation between the memory consolidation and sleep parameters.^{53,54}

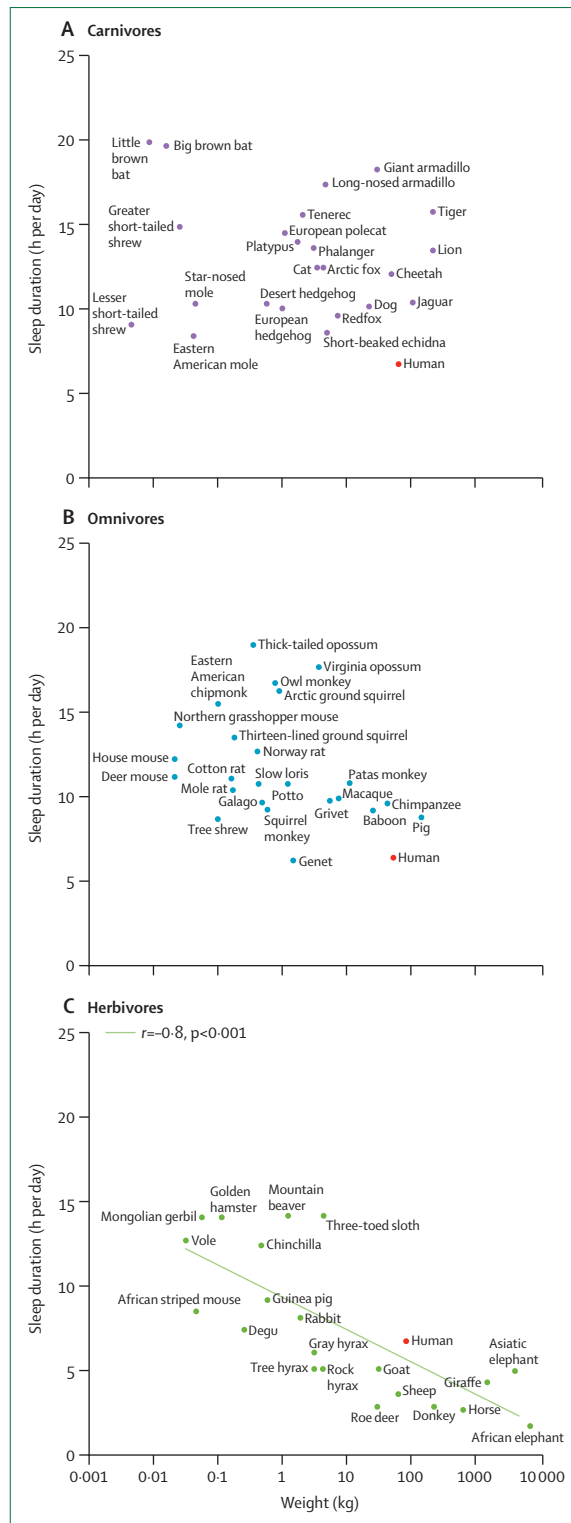
REM=rapid eye movement.

omnivores sleep more than herbivores. Humans, if categorised as carnivores, have the lowest sleep amount of any studied carnivore and, if categorised as omnivores, have the second lowest sleep duration of omnivores (only the common genet [*Viverra zibetha*], a type of cat, has less sleep). However, if categorised as herbivores, humans are almost exactly on the line of inverse correlation between body mass and sleep duration, suggesting that reliance on plant consumption had an important role in the evolution of human sleep duration. No evidence is available to show that the herbivore versus carnivore diet itself has a major effect on sleep amount (ie, there is no evidence that human vegetarians sleep for a shorter duration than do human carnivores or omnivores).

Contrary to the idea that sleep amounts are fixed, sleep duration can vary within species because of factors such as environmental temperature, migration periods, and mating behaviour. For example, reindeer (*Rangifer tarandus*) living in the arctic are active for 43% more of the 24 h period in summer (when food is available) than in winter (when snow and ice cover most of the edible

Figure 1: Sleep duration in mammals

Graphs show the sleep duration of carnivores, omnivores, and herbivores. Sleep data are compiled from various studies, many of which were done in zoos or laboratories.³⁸ The red dot highlights data for humans on each graph. Weight is plotted on a logarithmic scale. Log values for weight were used in correlation calculations versus linear values for sleep duration. The sleep durations of carnivores, omnivores, and herbivores differ significantly ($p < 0.0002$; f test). The sleep durations of carnivores are significantly greater than those of herbivores ($p < 0.001$; t test). Sleep amounts in herbivores are negatively correlated with weight ($r = -0.8$; $p < 0.001$). However, the correlation between weight and sleep time is not significant in carnivores ($r = -0.3$) or omnivores ($r = -0.3$).



See Online for appendix

vegetation).⁵⁹ Migrating animals, such as the white-crowned sparrow (*Zonotrichia leucophrys*), show a profound suppression of sleep during migratory periods. Even when some migrating birds are caged in captivity, sleep is greatly

reduced and activity is increased during seasons when they would be migrating in the wild, a phenomenon known as *zugunruhe* in German. This reduction in sleep is not associated with any decrement in learning ability, nor is it followed by a sleep rebound (ie, increased sleep after the end of the migratory period to make up for lost sleep during the migratory period).¹⁹

African bush elephants (*Loxodonta africana*) in the wild average only 2.1 h of sleep per day, which is the shortest amount of sleep reported in any mammal, and spend nearly all their waking hours eating or walking to find suitable vegetation.⁶⁰ The sleep duration of wild African elephants is half the time reported for elephants in captivity, who are fed daily and have an average 4–5 h sleep.³⁸ Elephants have the largest brain of any land mammal. Elephants also have a complex social structure, a cognitive ability rivalling that of primates, and one of the longest mammalian lifespans,³⁷ which is considerably longer for elephants in the wild than for those in captivity.⁶¹

The animal with the longest documented sleep time is the little brown bat (*Myotis lucifugus*). This animal sleeps, on average, 20 h per day.^{39,62} The little brown bat consumes moths and mosquitoes; these insects are in their nests during much of the daytime, become active at dusk, and return to their nests as the night progresses and environmental temperature drops. If the little brown bat awakened during the daytime, not only would their prey not be as readily available but also they would be subject to predation by carnivorous birds (eg, hawks and eagles). If the bats stayed out later at night, less food would be available and the cool evening would require greater energy expenditure. Thus, 20 h of daily sleep for the little brown bat is probably not a consequence of needing increased sleep for cognitive processing, but rather a consequence of these bats adapting to their ecological niche, in which their prey is available for a short daily period and their sleeping sites on cave walls or trees are relatively safe.

The functions of REM sleep

REM sleep was discovered in humans by Aserinsky and Kleitman,⁶³ and first recorded in an animal (the domestic cat [*Felis catus*]) by Dement.⁶⁴ REM sleep is seen in most homeotherm species, which include mammals and birds. Mammals can be divided into three subgroups: monotremes (ie, egg-laying mammals, such as the short-beaked echidna [*Tachyglossus aculeatus*] and the platypus [*Ornithorhynchus anatinus*]); marsupials (ie, mammals with pouches for their young, such as the North American opossum [*Lutreolina crassicaudata*], the thick-tailed opossum [*Lutreolina crassicaudata*], and the phalanger [*Trichosurus vulpecula*]); and placentals (ie, mammals that carry their fetus in a uterus, such as whales [Cetacea], elephants [Elephantidae], and humans; appendix). The first study of sleep in a monotreme, the short-beaked echidna, tentatively concluded that this species does not have REM sleep because the studied animals did not show

low voltage, cortical EEG during sleep. This feature is characteristic of REM sleep in adult placental mammals.⁶⁵ This tentative conclusion would accord with the idea that REM sleep is a phylogenetically new sleep state. However, a subsequent study of sleep in the short-beaked echidna found that, although the cortex shows high-voltage EEG, resembling that of non-REM sleep, the brainstem reticular formation shows burst-pause patterns of neuronal activity, closely resembling the pattern seen in REM sleep, throughout the echidna's sleep period.⁶⁶ In a subsequent study, sleep was recorded in another monotreme, the platypus (*Ornithorhynchus anatinus*).⁶⁷ REM sleep was observed in the platypus throughout much of the sleep period, although it often occurred during periods in which cortical EEG resembled the high-voltage patterns of non-REM sleep in placental and marsupial mammals, and in birds. This EEG pattern is similar to that recorded in the short-beaked echidna and to that of many newborn placental mammals (including humans). High-voltage cortical EEG is seen during so-called active sleep, which is the developmental precursor to REM sleep. The platypus has more REM sleep than any other animal, and the eye movements are typically accompanied by vigorous jaw and head twitching (video), which are similar to movements that occur in placental mammals during REM sleep. The

studies in the short-beaked echidna and platypus concluded that, contrary to the initial hypothesis, the REM sleep state might be the phylogenetically oldest sleep state in mammals. Monotremes belong to the ancestral line that gave rise to placentals and marsupials.

Three important points should be taken into consideration for understanding the function of REM sleep.⁵⁰ First, humans do not have unusually high or low amounts of REM sleep compared with other mammals (figure 2). Second, no obvious relation has been noted between REM sleep duration and presumed cognitive function. For example, the guinea pig (*Cavia porcellus*) has the same sleep parameters as does the guinea baboon (*Papio papio*) and the platypus and little brown bat are not known to have exceptional cognitive capabilities compared with humans and baboons. Third, the amount of REM sleep correlates with the amount of non-REM sleep across mammals (figure 3).

Brain temperature during non-REM sleep is lower than that during waking, but during REM sleep it rises to the brain temperature of waking.⁶⁸ During REM sleep, brainstem neuronal activity is greatly increased.⁴³ Monotremes are thought to have been transitional species between their ectothermic (ie, relying on the external environment to regulate body temperature) reptilian

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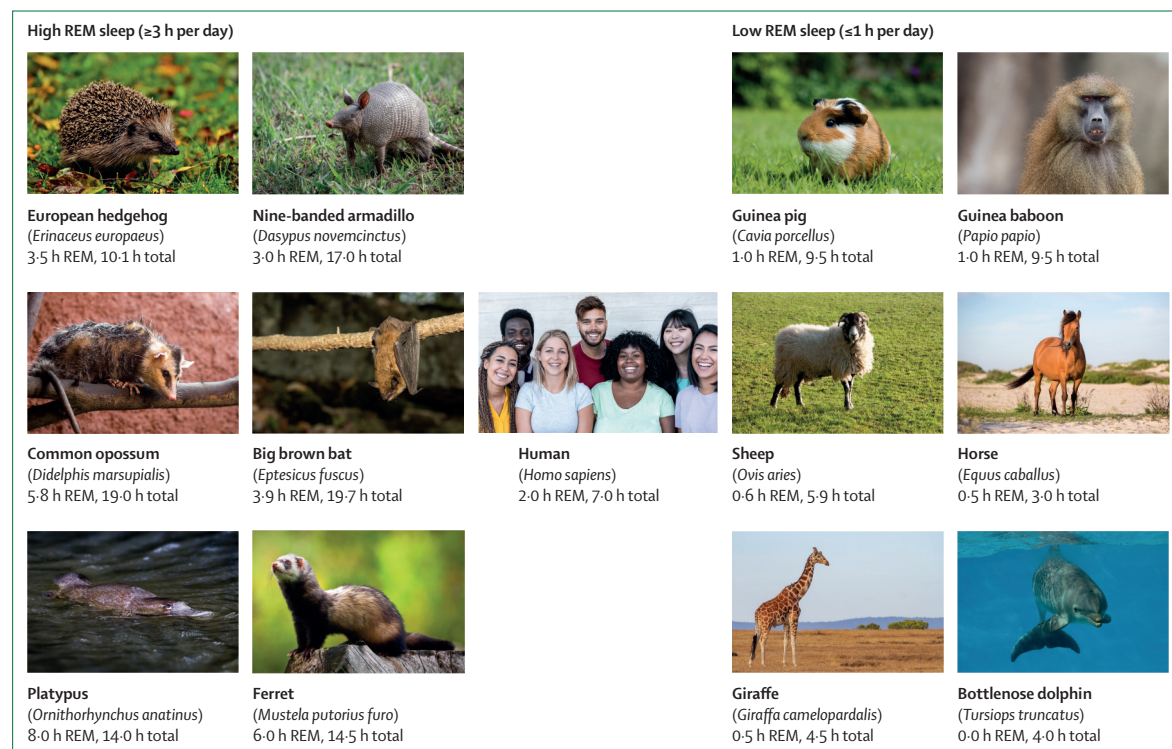


Figure 2: REM sleep duration across species

REM sleep time and total sleep time are shown for a selection of mammalian species. In contrast to other placental mammals, dolphins and other cetaceans only have unihemispheric sleep, so sleep time in dolphins is presented in this figure as half of the total 8 h unihemispheric sleep time. This figure was adapted from reference 50, with permission of the American Association for the Advancement of Science. REM=rapid eye movement. Photo credits: platypus, Martin Pelanek/Shutterstock; opossum, Vladislav T Jirousek/Shutterstock; ferret, jurra8/Shutterstock; big brown bat, Jay Ondreicka/Shutterstock; hedgehog, Vy nguyen 2905/Shutterstock; armadillo, Marcelo Morena/Shutterstock; humans, DisobeyArt/Shutterstock; guinea pig, Birute Vijeikiene/Shutterstock; guinea baboon, Nagel Photography/Shutterstock; sheep, Dawn Quadling/Shutterstock; horse, Mary Swift/Shutterstock; giraffe, Nicola_K_photos/Shutterstock; dolphin, Luis Sejjido/Shutterstock.

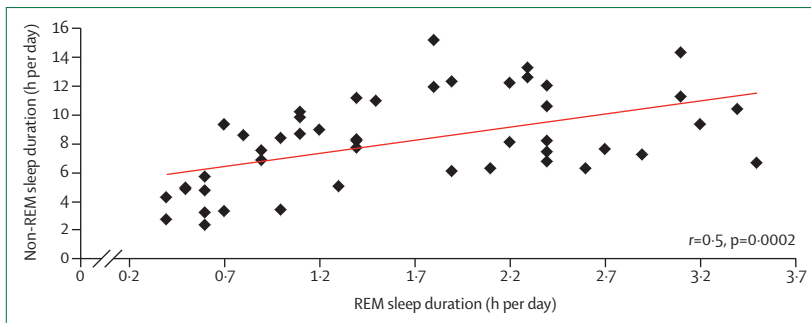


Figure 3: REM versus non-REM sleep duration across placental mammalian species
REM sleep duration is correlated with non-REM sleep duration across 47 placental mammalian species ($r=0.5$, $p=0.0002$). REM sleep and non-REM sleep data are shown in the appendix (pp 1–3). REM=rapid eye movement.

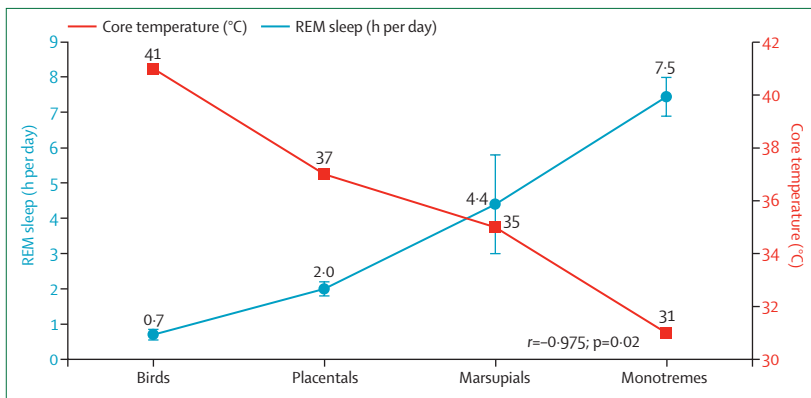


Figure 4: REM sleep hours per day versus core body temperature in homeotherm orders
Homeotherms with the highest body temperature (birds) have the lowest amount of REM sleep, with this inverse relation between temperature and REM sleep duration continuing across placental, marsupial, and monotreme species. Many bird and placental species have been studied, hence the very small standard error for their mean REM values. Overall, there is a negative correlation of core temperature with REM sleep duration across orders ($r=-0.975$; $p=0.02$).^{69–73} REM sleep data are shown in the appendix (pp 1–3). REM=rapid eye movement.

ancestors and endothermic (ie, generating heat to maintain a relatively fixed body temperature) mammals. The warming of the brain that occurs during REM sleep could explain the very long duration of REM sleep in monotremes (figure 4), which have lower body and brain temperatures than other mammals. Monotremes have a brain and body temperature of 31°C, whereas placental mammals have brain and body temperature of 37°C. Allowing brain temperature to fall lower than 31°C while asleep might interfere with autonomic function during sleep and with alert awakening, which is a characteristic of homeotherms that their ectothermic reptile ancestors do not have.^{38,69} The marsupial mammals that have been studied (eg, the common brushtail opossum [*Trichosurus vulpecula*], the North American opossum [*Didelphis virginiana*], and the big lutrine opossum [*Lutreolina crassicaudata*]) have a body temperature higher than that of monotremes but lower than that of placentals. The daily REM sleep duration for marsupials (about 4.4 h per day) is lower than that for monotremes (about 7.5 h per day), but higher than that for placentals (about 2.0 h per day). The placental mammal order has higher body temperatures and lower average daily REM sleep duration than marsupials. Within the

group of placental mammals, body temperatures are relatively fixed across species, and the ecological variables described in figure 1 are correlated with sleep duration. Birds have the highest body and brain temperatures of all homeotherm species (core temperature of 41°C), as well as the highest metabolic rates^{70,71} and the lowest daily amount of REM sleep (about 0.7 h per day—ie, a tenth of the amount of REM sleep in monotremes). Body and brain (including brainstem) temperature is a major (inverse) correlate of REM sleep time across homeotherm orders (figure 4).^{69–73}

REM sleep might have initially evolved to maintain brainstem function during sleep in mammals that had relatively low body temperatures and metabolic rates, which is a characteristic of the monotreme mammals. REM sleep might be thought of as a thermostatically controlled brain-heating mechanism, which is triggered by the temperature reduction linked to the reduced metabolism and the decrease in energy consumption in non-REM sleep.^{74,75} Then, REM sleep ends after the amount of REM required to raise brain temperature to close to the waking temperature of the body has occurred.

As marsupials and placentals with a higher body temperature and a proportionately larger forebrain have evolved, REM sleep activation has extended to the cortical regions, but its duration has decreased. The decerebrate mammal (ie, with the brainstem experimentally disconnected from the forebrain at the midbrain–forebrain junction) cannot thermoregulate. The body temperature of the decerebrate mammal, including the temperature of its blood, drifts towards room temperature if no external heat is applied (eg, through a heating pad). The pontine region of the brain is both necessary and sufficient for the generation of REM sleep, as revealed by systematic studies in domestic cats.^{76–78} Removing external heat from a decerebrate animal and allowing body temperature to fall to 23°C (ie, to room temperature) increases REM sleep duration, with REM sleep reaching 70% of the recording time (figure 5)⁷⁹ and REM sleep periods lasting for as long as 45 min.⁸⁰ This finding illustrates the thermosensitivity of the brainstem mechanisms generating REM sleep, with REM sleep being triggered by low brainstem temperatures, and increasing the temperature of the brain. By contrast, intact cats (ie, domestic cats that have not had their brainstem disconnected) have, on average, REM sleep for 15.5% of total sleep duration at thermoneutral temperatures,⁸¹ which is commensurate with the fact that intact cats have a smaller reduction in brain temperature during non-REM sleep than decerebrate cats. The intact cat never shows the long duration of REM sleep periods seen in the cold decerebrate cat.

These data together suggest that REM sleep has a thermoregulatory role for the brain, which is triggered by decreases in brainstem temperature during non-REM sleep, and ceases when the target temperature is reached, in a repeating cycle. Because the decerebrate animal cannot regulate its body temperature, the body and brain

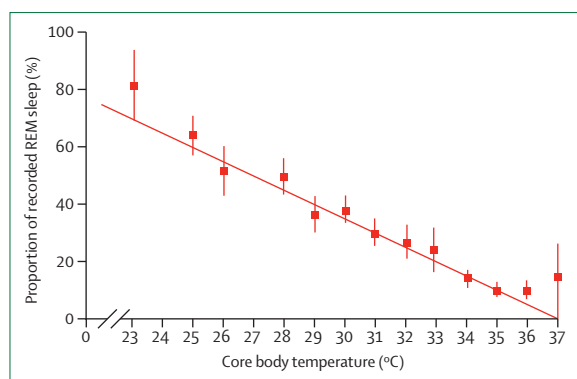


Figure 5: Effect of brain temperature on REM sleep

The decerebrate animal (ie, with the brainstem experimentally disconnected from the forebrain at the midbrain–forebrain junction) cannot regulate body temperature, which then falls towards room temperature. Data from decerebrate domestic cats (*Felis catus*)⁷⁸ show the effect of passive cooling of the pontine brainstem region, the region which is both necessary and sufficient for REM sleep, on the proportion of REM sleep. As core temperature falls, REM sleep increases well beyond the highest amount of REM sleep seen in the intact cat. Vertical lines represent standard errors, and the boxes denote the mean value. Graph drawn from data in reference 80. REM=rapid eye movement.

temperatures (including brainstem temperature) fall precipitously when external heating is removed, despite REM sleep activation. Therefore, REM sleep occurs nearly continuously. It could be said that REM sleep is like shivering for the brain.⁸²

In intact mammals (ie, mammals without a brain lesion or disconnection), REM sleep is normally triggered after non-REM sleep, which prevents brain temperature from falling far below the temperature required for alert waking on arousal. This occurrence fits with the known increase in REM sleep toward the end of the human sleep cycle,⁴ at a time when environmental and body temperature is generally low² and waking is imminent. The reduced body temperature during non-REM sleep provides a major energy benefit, because a large percentage of our energy consumption at rest is used to maintain body and brain temperature.³⁵ REM sleep periods typically follow non-REM sleep periods, consistent with a cycle of brain cooling during non-REM sleep followed by brain warming and correlated increases in brain metabolic rate,^{68,83,84} which are optimal conditions for alert awakening.

Body thermoregulation is maintained during non-REM sleep, albeit at a lower temperature than during waking. In REM sleep, brain temperature is increased from the temperature that occurs during non-REM sleep but body muscle tone and thermoregulation are greatly diminished. Body temperature will either increase or decrease depending on environmental temperature.⁸⁴ This combination of thermoregulatory changes that occur during REM sleep serves three functions. First, heating the brain keeps it metabolically functional and able to rapidly awaken. Second, reducing or eliminating activity in the muscles of the body saves energy. Third, reducing muscle tone prevents dream-enacting behaviours that are generated by pontine activation in REM sleep.⁸⁵ When the

Panel 3: REM and non-REM sleep pathologies

- Muscle tone is moderately reduced in non-REM sleep but is actively suppressed during REM sleep; this suppression is accomplished via inhibition by GABA and glycine release onto motor neurons, along with disfacilitation by reduction in monoamine release onto these neurons;^{85–87} this reduction in skeletal motor activity prevents the sleep disruption that would result from the very high level of activity of pontine brainstem movement-related neurons in REM sleep.^{47,85–87}
- Bilateral damage to the brainstem motor suppression systems, particularly to regions that are just ventral to the locus coeruleus,⁸⁵ results in REM sleep without atonia, in which extensive motor activity occurs during REM sleep; REM sleep behaviour disorder can occur in humans and causes apparent dream-enacting behaviours; in contrast to the localised lesion in this animal model, REM sleep behaviour disorder is a synucleinopathy, with no discretely localised lesion; more than 80% of individuals with REM sleep behaviour disorder go on to develop Parkinson's disease.^{87,88}
- In obese individuals with narrowed airways, and in some individuals with small airways and normal weight, the suppression of pharyngeal muscle tone in REM sleep can cause an occlusion of the airway. Occlusion of the airway can result in sleep apnoea, cyclically alternating with arousals, leading to sleep fragmentation, oxyhaemoglobin desaturation, hypertension, cardiovascular disease, and other related problems; sleep apnoea can be successfully treated with continuous positive airway pressure applied through a facial mask or directly through the nose.⁸⁹
- The pathology underlying the common syndrome of periodic movements during sleep in individuals with restless legs syndrome remains to be elucidated, although iron deficiency and dopamine appear to have central roles.⁹⁰
- Narcolepsy is caused by a loss of hypocretin neurons^{91,94} and can be associated with psychological depression and loss of muscle tone with emotions in waking (cataplexy);^{92,93} this sleep disorder is discussed in more detail in the second paper in *The Lancet Series on sleep*, which is on excessive daily sleepiness.¹⁴

REM=rapid eye movement.

muscle tone suppression mechanism of REM sleep is damaged, REM sleep behaviour disorder occurs (panel 3).^{86–94}

REM sleep can be pharmacologically suppressed in humans for weeks to months by monoamine oxidase inhibitors such as phenelzine.⁹⁵ No periods of rapid eye movement occur, no periods of REM sleep-like EEG activation occur, and no dreams are reported.⁹⁵ This suppression is not accompanied by any detected disturbance of cognitive function. The reduction of REM sleep by serotonin and norepinephrine reuptake inhibitors also has no negative effects on memory.⁹⁶ No relation has been noted between measures of human intelligence and REM sleep duration.⁹⁷ Moreover, in a rare case, an individual who survived damage to the pontine region, which is known to generate REM sleep, showed average or above average learning and cognition on various tests despite minimal amounts of REM sleep.⁹⁸

Non-REM sleep and REM sleep in marine mammals

The adaptations that have evolved in marine mammals provide important insights into the functional roles of mammalian sleep states. Extant marine mammals have

Panel 4: Inactive states in non-homeothermic species

Some cicada species are successful because they limit activity to a 2–4 week period, then become inactive for as long as 17 years.¹⁰³ A similar phenomenon has been reported in millipedes.¹⁰⁴ Small fruit flies (*Drosophila melanogaster*),¹⁰⁵ free-living nematodes (*Caenorhabditis elegans*),¹⁰⁶ and zebrafish (*Danio rerio*)¹⁰⁷ have states that have been described as sleep-like. However, the differences between mammalian brains and the neuronal organisation of these poikilothermic organisms make functional comparisons difficult. In the only study of brainstem neuronal activity during sleep in a reptile,¹⁰⁸ no evidence was found for a REM sleep-like state. It cannot be assumed that all states of reduced behavioural activity are homologous to mammalian sleep states.

Search strategy and selection criteria

I searched MEDLINE and Google with the keywords (“sleep” AND “neurons”), (“sleep” AND “cognition” OR “learning”), “sleep function”, (“sleep” AND “mammals”), and “sleep phylogeny”. The search was restricted to publications from Jan 1, 2000, to June 1, 2021, with no language restrictions. I also included key references from before 2000.

very large brains and high brain-to-bodyweight ratios, which are almost as high as the brain-to-bodyweight ratios seen in primates.^{99,100} Common bottlenose dolphins (*Tursiops truncatus*) have brains that are substantially larger than human brains. However, instead of the bilateral slow waves seen during sleep in land mammals, these dolphins only have unihemispheric slow waves during sleep. This type of sleep has been described as unihemispheric sleep or half brain sleep. In contrast to all previously studied mammals, no evidence has been found that dolphins have REM sleep (figure 2), with similar findings in other cetacean species.⁹⁹

Because most land mammals have their longest durations of REM sleep shortly after birth, with a reduced duration of REM sleep in adults, the idea was proposed that newborn dolphins might show REM sleep, which would be consistent with the hypothesis that REM sleep plays an essential role in neural development. However, not only do newborn dolphins and killer whales (*Orcinus orca*), which are the largest species of the dolphin family, not show REM sleep, but they do not sleep at all for several weeks after birth.¹⁰¹ In the wild, killer whale calves and their mothers (and other cetaceans) often migrate for thousands of miles after birth to get from their birthing area to feeding areas. This migration requires high levels of alertness and sensory motor control, because the calves are subject to predation by sharks and other killer whales, which means that the mother and calf must stay together and be responsive to threats.

The absence of REM sleep in dolphins was put into a larger context by work in fur seals (*Callorhinus ursinus*).¹⁰² Fur seals have about 80 min of REM sleep per day and about 5–6 h of non-REM sleep per day when they sleep on land, with bilateral high-voltage cortical activity, similar to the sleep amounts in humans. When fur seals go into

water, which is their home for about 7 months of the year, they do not show the bilateral non-REM sleep that they have when on land. Instead, fur seals show unihemispheric high-voltage activity, similar to that seen in the dolphin and other cetacean species. Fur seals also largely cease having REM sleep when they are in water. Moreover, when fur seals return to land, they do not show the large increase in non-REM and REM sleep that can be seen after sleep deprivation in terrestrial mammals.¹⁰²

Unlike bilateral non-REM sleep, unihemispheric sleep might not cool the brainstem. This absence of cooling could enable cetaceans to maintain coordinated brainstem motor and sensory function, despite unihemispheric cortical slow waves. Because of the absence of cooling, warming of the brainstem with REM sleep would be unnecessary. This hypothesis could be tested in fur seals by comparing their brainstem temperatures during sleep on land with brainstem temperatures during unihemispheric sleep when they are in the water environment.

Non-homeothermic animals (ie, poikilothermic species such as reptiles and insects) vary their activity over the 24 h cycle. Inactive states in these creatures might serve the same energy conservation function as sleep does in homeotherms. However, whether these inactive states are homologous (ie, coming from a common evolutionary root) or analogous (ie, serving a similar function by using an unrelated biological mechanism) is unclear (panel 4).^{103–108}

Conclusions and future directions

Sleep is a highly adaptive brain function because it reduces energy expenditure. Insomnia is common in humans living in industrial societies, which is probably due, at least to some extent, to the removal of natural temperature and light cycles from the modern environment. The means by which environmental variables affect neurons regulating sleep remain to be determined. Body and pontine brainstem temperature recordings should be done during animal sleep recordings to better understand the relation of sleep to thermoregulation.

Declaration of interests

I declare no competing interests.

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References

- McNew JJ, Burson RC, Hoshizaki T, Adey WR. Sleep-wake cycle of an unrestrained isolated chimpanzee under entrained and free running conditions. *Aerosp Med* 1972; **43**: 155–61.
- Yetish G, Kaplan H, Gurven M, et al. Natural sleep and its seasonal variations in three pre-industrial societies. *Curr Biol* 2015; **25**: 2862–68.
- Youngstedt SD, Goff EE, Reynolds AM, et al. Has adult sleep duration declined over the last 50+ years? *Sleep Med Rev* 2016; **28**: 69–85.
- Carskadon MA, Dement WC. Normal human sleep. In: Kryger MH, Roth T, Dement WC, eds. *Principles and practice of sleep medicine*, 4th edn. Philadelphia, PA: W.B. Saunders, 2005: 13–23.
- Kingsbury JH, Buxton OM, Emmons KM, Redline S. Sleep and its relationship to racial and ethnic disparities in cardiovascular disease. *Curr Cardiovasc Risk Rep* 2013; **7**: 10.

- 6 Ayas NT, White DP, Manson JE, et al. A prospective study of sleep duration and coronary heart disease in women. *Arch Intern Med* 2003; **163**: 205–09.
- 7 Chen Y, Tan F, Wei L, et al. Sleep duration and the risk of cancer: a systematic review and meta-analysis including dose-response relationship. *BMC Cancer* 2018; **18**: 1149.
- 8 Kripke DF. Chronic hypnotic use: deadly risks, doubtful benefit: REVIEW ARTICLE. *Sleep Med Rev* 2000; **4**: 5–20.
- 9 Kripke DF, Garfinkel L, Wingard DL, Klauber MR, Marler MR. Mortality associated with sleep duration and insomnia. *Arch Gen Psychiatry* 2002; **59**: 131–36.
- 10 Partinen M, Putkonen PTS, Kaprio J, Koskenvuo M, Hilakivi I. Sleep disorders in relation to coronary heart disease. *Acta Med Scand Suppl* 1982; **660**: 69–83.
- 11 Li Y, Sahakian BJ, Kang J, et al. The brain structure and genetic mechanisms underlying the nonlinear association between sleep duration, cognition and mental health. *Nature Aging* 2022; **2**: 425–37.
- 12 Warburton DER, Nicol CW, Bredin SSD. Health benefits of physical activity: the evidence. *CMAJ* 2006; **174**: 801–09.
- 13 Zitter J, Anaturk M, Zsoldos E, et al. Sleep duration over 28 years, cognition, gray matter volume, and white matter microstructure: a prospective cohort study. *Sleep* 2020; **43**: 1–7.
- 14 Pérez-Carbonell L, Mignot E, Leschziner G, Dauvilliers Y. Understanding and approaching excessive daytime sleepiness. *Lancet* 2022; published online Sept 14. [https://doi.org/10.1016/S0140-6736\(22\)01018-2](https://doi.org/10.1016/S0140-6736(22)01018-2).
- 15 Perlis ML, Posner D, Riemann D, Bastien CH, Tell J, Thase M. Insomnia. *Lancet* 2022; published online Sept 14. [https://doi.org/10.1016/S0140-6736\(22\)00879-0](https://doi.org/10.1016/S0140-6736(22)00879-0).
- 16 Meyer N, Harvey AG, Lockley SW, Dijk D-J. Circadian rhythms and disorders of the timing of sleep. *Lancet* 2022; published online Sept 14. [https://doi.org/10.1016/S0140-6736\(22\)00877-7](https://doi.org/10.1016/S0140-6736(22)00877-7).
- 17 Santhi N, Thorne HC, van der Veen DR, et al. The spectral composition of evening light and individual differences in the suppression of melatonin and delay of sleep in humans. *J Pineal Res* 2012; **53**: 47–59.
- 18 Davimes JG, Alagaili AN, Bhagwandin A, et al. Seasonal variations in sleep of free-ranging Arabian oryx (*Oryx leucorox*) under natural hyperarid conditions. *Sleep* 2018; **41**: zsy038.
- 19 Rattenborg NC, Mandt BH, Obermeyer WH, et al. Migratory sleeplessness in the white-crowned sparrow (*Zonotrichia leucophrys gambelii*). *PLoS Biol* 2004; **2**: E212.
- 20 van Oort BE, Tyler NJ, Gerkema MP, Folkow L, Blix AS, Stokkan KA. Circadian organization in reindeer. *Nature* 2005; **438**: 1095–96.
- 21 Buysse DJ. Insomnia. *JAMA* 2013; **309**: 706–16.
- 22 Roth T. Insomnia: definition, prevalence, etiology, and consequences. *J Clin Sleep Med* 2007; **3** (suppl): S7–10.
- 23 Watson NF, Badr MS, Belenky G, et al. Recommended amount of sleep for a healthy adult: a joint consensus statement of the American Academy of Sleep Medicine and Sleep Research Society. *J Clin Sleep Med* 2015; **11**: 591–92.
- 24 Kaplan H, Thompson RC, Trumble BC, et al. Coronary atherosclerosis in indigenous South American Tsimane: a cross-sectional cohort study. *Lancet* 2017; **389**: 1730–39.
- 25 Gurven M, Blackwell AD, Rodríguez DE, Stieglitz J, Kaplan H. Does blood pressure inevitably rise with age?: longitudinal evidence among forager-horticulturalists. *Hypertension* 2012; **60**: 25–33.
- 26 Trumble BC, Stieglitz J, Rodríguez DE, Linares EC, Kaplan HS, Gurven MD. Challenging the inevitability of prostate enlargement: low levels of benign prostatic hyperplasia among Tsimane forager-horticulturalists. *J Gerontol A Biol Sci Med Sci* 2015; **70**: 1262–68.
- 27 Trumble BC, Stieglitz J, Blackwell AD, et al. Apolipoprotein E4 is associated with improved cognitive function in Amazonian forager-horticulturalists with a high parasite burden. *FASEB J* 2017; **31**: 1508–15.
- 28 Zeng W, Eisenberg DT, Jovel KR, et al. Adult obesity: panel study from native Amazonians. *Econ Hum Biol* 2013; **11**: 227–35.
- 29 Pisor AC, Gurven M, Blackwell AD, Kaplan H, Yetish G. Patterns of senescence in human cardiovascular fitness: VO2 max in subsistence and industrialized populations. *Am J Hum Biol* 2013; **25**: 756–69.
- 30 Pontzer H, Wood BM, Raichlen DA. Hunter-gatherers as models in public health. *Obes Rev* 2018; **19** (suppl 1): 24–35.
- 31 Moss M. Hooked: food, free will, and how the food giants exploit our addictions. New York, NY: Random House, 2021.
- 32 Lagranja ES, Phojanakong P, Navarro A, Valeggia CR. Indigenous populations in transition: an evaluation of metabolic syndrome and its associated factors among the Toba of northern Argentina. *Ann Hum Biol* 2015; **42**: 84–90.
- 33 von der Ohe CG, Garner CC, Darian-Smith C, Heller HC. Synaptic protein dynamics in hibernation. *J Neurosci* 2007; **27**: 84–92.
- 34 Wolf BO, McKechnie AE, Schmitt CJ, Czenze ZJ, Johnson AB, Witt CC. Extreme and variable torpor among high-elevation Andean hummingbird species. *Biol Lett* 2020; **16**: 20200428.
- 35 Schmidt MH. The energy allocation function of sleep: a unifying theory of sleep, torpor, and continuous wakefulness. *Neurosci Biobehav Rev* 2014; **47**: 122–53.
- 36 Nofzinger EA, Nissen C, Germain A, et al. Regional cerebral metabolic correlates of WASO during NREM sleep in insomnia. *J Clin Sleep Med* 2006; **2**: 316–22.
- 37 Poole J, Granli P. Mind and movement: meeting the interests of elephants. In: Forthman DL, Kane LF, Waldau PF, Atkinson RPD, eds. An elephant in the room: the science and well-being of elephants in captivity. Medford MA: Tufts Center for Animals & Public Policy; 2008: 1–21.
- 38 Siegel JM. Clues to the functions of mammalian sleep. *Nature* 2005; **437**: 1264–71.
- 39 Siegel JM. Sleep viewed as a state of adaptive inactivity. *Nat Rev Neurosci* 2009; **10**: 747–53.
- 40 Siegel JM. Evolution of mammalian sleep. In: Kryger MK, Roth T, Goldstein CA, Dement WC, eds. Principles and practice of sleep medicine, 7th edn. Amsterdam: Elsevier; 2022: 93–105.
- 41 Siegel JM. Introduction/Defining sleep. In: Kryger MH, Roth T, Goldstein CA, Dement WC, eds. Principles and practice of sleep medicine, 7th edn. Amsterdam: Elsevier; 2022: 52–53.
- 42 Xie L, Kang H, Xu Q, et al. Sleep drives metabolite clearance from the adult brain. *Science* 2013; **342**: 373–77.
- 43 Siegel JM. Rapid eye movement sleep control and function. In: Kryger MH, Roth T, Goldstein CA, Dement WC, eds. Principles and practice of sleep medicine, 7th edn. Amsterdam: Elsevier; 2022: 68–86.
- 44 Cirelli C, Tononi G. Effects of sleep and waking on the synaptic ultrastructure. *Philos Trans R Soc Lond B Biol Sci* 2020; **375**: 20190235.
- 45 Clow A, Law R, Evans P, et al. Day differences in the cortisol awakening response predict day differences in synaptic plasticity in the brain. *Stress* 2014; **17**: 219–23.
- 46 Kastaun S, Schwarz NP, Juenemann M, et al. Cortisol awakening and stress response, personality and psychiatric profiles in patients with Takotsubo cardiomyopathy. *Heart* 2014; **100**: 1786–92.
- 47 Siegel JM. Behavioral functions of the reticular formation. *Brain Res* 1979; **180**: 69–105.
- 48 Ramanathan L, Gulyani S, Nienhuis R, Siegel JM. Sleep deprivation decreases superoxide dismutase activity in rat hippocampus and brainstem. *Neuroreport* 2002; **13**: 1387–90.
- 49 Rechtschaffen A, Bergmann BM. Sleep deprivation in the rat: an update of the 1989 paper. *Sleep* 2002; **25**: 18–24.
- 50 Siegel JM. The REM sleep-memory consolidation hypothesis. *Science* 2001; **294**: 1058–63.
- 51 Vertes RP, Linley SB. No cognitive processing in the unconscious, anesthetic-like, state of sleep. *J Comp Neurol* 2021; **529**: 524–38.
- 52 Siegel JM. Memory consolidation is similar in waking and sleep. *Curr Sleep Med Rep* 2021; **7**: 15–18.
- 53 Cordi MJ, Rasch B. How robust are sleep-mediated memory benefits? *Curr Opin Neurobiol* 2021; **67**: 1–7.
- 54 Cordi MJ, Rasch B. No evidence for intra-individual correlations between sleep-mediated declarative memory consolidation and slow-wave sleep. *Sleep* 2021; **44**: 1–10.
- 55 Craig M, Dewar M, Della Sala S, Wolbers T. Rest boosts the long-term retention of spatial associative and temporal order information. *Hippocampus* 2015; **25**: 1017–27.
- 56 Humiston GB, Tucker MA, Summer T, Wamsley EJ. Resting states and memory consolidation: a preregistered replication and meta-analysis. *Sci Rep* 2019; **9**: 19345.
- 57 Ackermann S, Hartmann F, Papassotiropoulos A, de Quervain DJ, Rasch B. No associations between interindividual differences in sleep parameters and episodic memory consolidation. *Sleep* 2015; **38**: 951–59.

- 58 Zepelin H. Mammalian sleep. In: Kryger MH, Roth T, Dement WC, eds. Principles and practice of sleep medicine. Philadelphia, PA: WB Saunders, 2000: 82–92.
- 59 van Oort BE, Tyler NJ, Gerkema MP, Folkow L, Stokkan KA. Where clocks are redundant: weak circadian mechanisms in reindeer living under polar photic conditions. *Naturwissenschaften* 2007; **94**: 183–94.
- 60 Gravett N, Bhagwandin A, Sutcliffe R, et al. Inactivity/sleep in two wild free-roaming African elephant matriarchs—does large body size make elephants the shortest mammalian sleepers? *PLoS One* 2017; **12**: e0171903.
- 61 Clubb R, Rowcliffe M, Lee P, Mar KU, Moss C, Mason GJ. Compromised survivorship in zoo elephants. *Science* 2008; **322**: 1649.
- 62 Harding CD, Yovel Y, Peirson SN, Hackett TD, Vyazovskiy VV. Re-examining extreme sleep duration in bats: implications for sleep phylogeny, ecology and function. *Sleep* 2022; published online March 13. <https://doi.org/10.1093/sleep/zsac064>.
- 63 Aserinsky E, Kleitman N. Regularly occurring periods of eye motility, and concomitant phenomena, during sleep. *Science* 1953; **118**: 273–74.
- 64 Dement W. The occurrence of low voltage, fast, electroencephalogram patterns during behavioral sleep in the cat. *Electroencephalogr Clin Neurophysiol* 1958; **10**: 291–96.
- 65 Allison T, Van Twyver H, Goff WR. Electrophysiological studies of the echidna, *Tachyglossus aculeatus*. I. Waking and sleep. *Arch Ital Biol* 1972; **110**: 145–84.
- 66 Siegel JM, Manger PR, Nienhuis R, Fahringer HM, Pettigrew JD. The echidna *Tachyglossus aculeatus* combines REM and non-REM aspects in a single sleep state: implications for the evolution of sleep. *J Neurosci* 1996; **16**: 3500–06.
- 67 Siegel JM, Manger PR, Nienhuis R, Fahringer HM, Shalita T, Pettigrew JD. Sleep in the platypus. *Neuroscience* 1999; **91**: 391–400.
- 68 Parmeggiani PL. REM sleep related increase in brain temperature: a physiologic problem. *Arch Ital Biol* 2007; **145**: 13–21.
- 69 Tan CL, Knight ZA. Regulation of body temperature by the nervous system. *Neuron* 2018; **98**: 31–48.
- 70 Clarke A, Rothery P. Scaling of body temperature in mammals and birds. *Funct Ecol* 2008; **22**: 58–67.
- 71 Roth TC 2nd, Lesku JA, Amlaner CJ, Lima SL. A phylogenetic analysis of the correlates of sleep in birds. *J Sleep Res* 2006; **15**: 395–402.
- 72 Clarke A, Rothery P, Isaac NJB. Scaling of basal metabolic rate with body mass and temperature in mammals. *J Anim Ecol* 2010; **79**: 610–19.
- 73 White CR, Seymour RS. Mammalian basal metabolic rate is proportional to body mass^{2/3}. *Proc Natl Acad Sci USA* 2003; **100**: 4046–49.
- 74 McGinty D, Szymusiak R. Keeping cool: a hypothesis about the mechanisms and functions of slow-wave sleep. *Trends Neurosci* 1990; **13**: 480–87.
- 75 Wisor JP, Remppe MJ, Schmidt MA, Moore ME, Clegern WC. Sleep slow-wave activity regulates cerebral glycolytic metabolism. *Cereb Cortex* 2013; **23**: 1978–87.
- 76 Jouvet M. Recherches sur les structures nerveuses et les mécanismes responsables des différentes phases du sommeil physiologique. *Arch Ital Biol* 1962; **100**: 125–206 (in French).
- 77 Siegel JM, Nienhuis R, Tomaszewski KS. REM sleep signs rostral to chronic transections at the pontomedullary junction. *Neurosci Lett* 1984; **45**: 241–46.
- 78 Siegel JM, Tomaszewski KS, Nienhuis R. Behavioral states in the chronic medullary and midpontine cat. *Electroencephalogr Clin Neurophysiol* 1986; **63**: 274–88.
- 79 Jouvet M, Buda C, Sastre JP. Is there a bulbar pacemaker responsible for the ultradian rhythm of paradoxical sleep? *Arch Ital Biol* 1995; **134**: 39–56 (in French).
- 80 Jouvet M, Buda C, Sastre JP. Progressive hypothermia is accompanied by an almost permanent paradoxical sleep in pontine cats. *Sleep Res* 1990; **19**: 38.
- 81 Stermann MB, Knauss T, Lehmann D, Clemente CD (1965). Circadian sleep and waking patterns in the laboratory cat. *Electroencephalogr Clin Neurophysiol* 1965; **19**: 509–17.
- 82 Wehr TA. A brain-warming function for REM sleep. *Neurosci Biobehav Rev* 1992; **16**: 379–97.
- 83 Ursin R. The two stages of slow wave sleep in the cat and their relation to REM sleep. *Brain Res* 1968; **11**: 347–56.
- 84 Parmeggiani PL. Thermoregulation and sleep. *Front Biosci* 2003; **8**: s557–67.
- 85 Jouvet M, Delorme F. Locus coeruleus et sommeil paradoxal. *C R Soc Biol* 1965; **159**: 895–99 (in French).
- 86 Lai YY, Kodama T, Siegel JM. Changes in monoamine release in the ventral horn and hypoglossal nucleus linked to pontine inhibition of muscle tone: an in vivo microdialysis study. *J Neurosci* 2001; **21**: 7384–91.
- 87 Lai YY, Kodama T, Schenkel E, Siegel JM. Behavioral response and transmitter release during atonia elicited by medial medullary stimulation. *J Neurophysiol* 2010; **104**: 2024–33.
- 88 Schenck CH. Rapid eye movement sleep behavior disorder: current knowledge and future directions. *Sleep Med* 2013; **14**: 699–702.
- 89 Sullivan CE. Nasal positive airway pressure and sleep apnea. Reflections on an experimental method that became a therapy. *Am J Respir Crit Care Med* 2018; **198**: 581–87.
- 90 Salminen AV, Silvani A, Allen RP, et al. Consensus guidelines on rodent models of restless legs syndrome. *Mov Disord* 2021; **36**: 558–69.
- 91 Thannickal TC, Moore RY, Nienhuis R, et al. Reduced number of hypocretin neurons in human narcolepsy. *Neuron* 2000; **27**: 469–74.
- 92 Lee MJ, Lee SY, Yuan SS, et al. Comorbidity of narcolepsy and depressive disorders: a nationwide population-based study in Taiwan. *Sleep Med* 2017; **39**: 95–100.
- 93 Nordstrand SEH, Hansen BH, Rootwelt T, et al. Psychiatric symptoms in patients with post-H1N1 narcolepsy type 1 in Norway. *Sleep* 2019; **42**: zsz008.
- 94 Peyron C, Faraco J, Rogers W, et al. A mutation in a case of early onset narcolepsy and a generalized absence of hypocretin peptides in human narcoleptic brains. *Nat Med* 2000; **6**: 991–97.
- 95 Wyatt RJ, Fram DH, Kupfer DJ, Snyder F. Total prolonged drug-induced REM sleep suppression in anxious-depressed patients. *Arch Gen Psychiatry* 1971; **24**: 145–55.
- 96 Rasch B, Pommer J, Diekelmann S, Born J. Pharmacological REM sleep suppression paradoxically improves rather than impairs skill memory. *Nat Neurosci* 2009; **12**: 396–97.
- 97 Borrow SJ, Adam K, Chapman K, Oswald I, Hudson L, Idzikowski CJ. REM sleep and normal intelligence. *Biol Psychiatry* 1980; **15**: 165–69.
- 98 Magidov E, Hayat H, Sharon O, et al. Near-total absence of REM sleep co-occurring with normal cognition: an update of the 1984 paper. *Sleep Med* 2018; **52**: 134–37.
- 99 Lyamin OI, Manger PR, Ridgway SH, Mukhametov LM, Siegel JM. Cetacean sleep: an unusual form of mammalian sleep. *Neurosci Biobehav Rev* 2008; **32**: 1451–84.
- 100 Lyamin OI, Siegel JM. Sleep in aquatic mammals. *Hand Behav Neurosci* 2019; **30**: 375–93.
- 101 Lyamin O, Pryaslova J, Lance V, Siegel J. Animal behaviour: continuous activity in cetaceans after birth. *Nature* 2005; **435**: 1177.
- 102 Lyamin OI, Kosenko PO, Korneva SM, Vyssotski AL, Mukhametov LM, Siegel JM. Fur seals suppress REM sleep for very long periods without subsequent rebound. *Curr Biol* 2018; **28**: 2000–2005.e2.
- 103 Ito H, Kakishima S, Uehara T, et al. Evolution of periodicity in periodical cicadas. *Sci Rep* 2015; **5**: 14094.
- 104 Nijijima K, Nii M, Yoshimura J. Eight-year periodical outbreaks of the train millipede. *R Soc Open Sci* 2021; **8**: 201399.
- 105 Donlea J, Leahy A, Thimman MS, et al. Foraging alters resilience/vulnerability to sleep disruption and starvation in *Drosophila*. *Proc Natl Acad Sci USA* 2012; **109**: 2613–18.
- 106 Lawler DE, Chew YL, Hawk JD, Aljohbe A, Schafer WR, Albrecht DR. Sleep analysis in adult *C. elegans* reveals state-dependent alteration of neural and behavioral responses. *J Neurosci* 2021; **41**: 1892–907.
- 107 Leung LC, Wang GX, Madeline R, et al. Neural signatures of sleep in zebrafish. *Nature* 2019; **571**: 198–204.
- 108 Eiland MM, Lyamin OI, Siegel JM. State-related discharge of neurons in the brainstem of freely moving box turtles, *Terrapene carolina major*. *Arch Ital Biol* 2001; **139**: 23–36.

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